Examiner reports

Q1.

Students seemed comfortable with the transformation required in this question with around 62% correctly choosing Figure 3. Around 23% mixed up the required stretch and incorrectly chose Figure 5.

Q2.

A high proportion of students selected the correct response for this question. Incorrect responses were split between the available options fairly evenly.

Q3.

Part (a) was well attempted and most students gained some credit. Too many students lost marks for poorly drawn curves, which had the wrong curvature, lacked symmetry or were excessively wobbly.

The most frequent mark for part (b) was 2/4. The question discriminated well between students of varying abilities. While most students correctly used the discriminant > 0 for two real roots, many overlooked the fact that the roots were both positive, so only found half of the solution. Few students achieved full marks as insufficient justification/explanation was seen.

Q4.

Many students made good progress with this question and chose an appropriate technique to be able to determine the transformations. Most identified $R \sin (x - \alpha)$ as the best approach, although some students chose $R \sin (x + \alpha)$ and a few individuals started with $R \sin (x \pm \alpha)$, which often resulted in sign errors.

Having decided on an approach, most students handled well the routine of finding R and α although some used some inefficient methods and did not seem sufficiently well practiced in what is a very standard technique.

In order to fully justify their answer a clear comparison of the given equation and their $R \sin (x \pm \alpha) + 4$ needed to be made and in a few cases students' reasoning was not clear. Most students achieved some credit for correctly describing transformations relating to their $R \sin (x \pm \alpha) + 4$. All was not lost if mistakes were made early in the solution.

Part (b)(i) hinged on students realising that the denominator had to be maximised and so their $\sin (x \pm \alpha)$ had to equal 1. Students who had gone wrong in part (a) could still achieve some credit in part (b)(i). Many students gave several lines of working to rationalise the denominator of their fraction. This was unnecessary given allowed calculator technology.

The mark for part (b)(ii) was available to students who had gone wrong in part (a). Provided they used their R correctly, work was followed through.

Q8.

(a) (i) Many students were successful in completing the square, although some gave their answer as $(x – 3)^2 + 20$ instead of $(x – 3)^2 + 2$.

(ii) Most students ignored the request in bold type and considered the discriminant of the quadratic. An algebraic approach was expected here, but few made much progress beyond writing $(x – 3)^2 = – 2$. It was necessary to reason that no real square root of –2 exists and therefore the equation has no real solutions. Other acceptable reasoning was based on the fact that the minimum value of the expression $(x – 3)^2 + 2$ was equal to 2.

(b) (i) The term “vertex” is becoming better understood and most students realised that they could write down its coordinates from the completed square form of the quadratic. A few used differentiation to find the stationary point, which was also acceptable.

(ii) Those with the correct vertex usually drew a correct parabola with the value of the intercept on the y-axis clearly shown or stated.

(iii) Many more students are now using the correct term “translation”, but many still used an incorrect word such as “move” or “shift” or also involved another transformation such as a “stretch”. The most common, but incorrect, vector stated by students was $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and this suggests that they had not read the question carefully enough.

Q9.

In part (a) a large proportion of the students recognised the transformation as a stretch although a significant minority incorrectly stated that it was in the x -direction with scale factor 1 / 3. In part (b) there were many correct sketches although it was not uncommon to see the
y – intercept incorrectly given as 1. In part (c), the majority of students scored the first mark by eliminating y correctly to obtain $4^{-x} = 3 \times 4^x$ although a significant minority failed to score even this opening mark because they tried to combine two steps which resulted in the incorrect initial equation, $-x \log 4 = x \log 12$. A significant number of the students failed to make any further progress. More able students who used the log laws correctly to reach $-x \log 4 = \log 3 + x \log 4$ were generally successful in finding the correct value for x. A common error in applying laws of indices, amongst those students who tried to rearrange the equation $4^{-x} = 3 \times 4^x$ before taking logarithms, is illustrated by the incorrect equation $16^{2x} = \frac{1}{3}$.

Q10.

This final question resulted in a wide spread of marks. In part (a) a significant number of students did not see that the answers could be written down directly. The most common incorrect method led to incorrect equations similar to $x + 30 = 89.27$, the right-hand side coming from $\tan^{-1}(79)$. Most students correctly recognised the geometrical transformation in part (b) to be a translation but the direction of the translation was quite frequently incorrect. In part (c)(i) the right-hand side of the given equation was usually expanded correctly with the correct notation seen and many recalled and used the correct relevant trigonometric identity to eliminate $\sin^2 \theta$. However, solutions towards the printed answer were spoilt in a significant minority of cases by students incorrectly writing the correct expression $5 + 1 - \cos^2\theta$ as $5 - 5\cos^2\theta$. Of those who obtained the correct simplified quadratic equation in $\cos\theta$, most were able to solve the equation correctly but not all of these students gave an indication that they had considered and rejected the value –2 for $\cos\theta$. The final part of this last question proved to be a challenge even for the more able students. Many solutions just consisted of solving $\cos\theta = \frac{3}{4}$ to obtain two solutions rather than stating the required equation as $\cos 2x\ \frac{3}{4}$ and solving it to find the four solutions within the given range to the required degree of accuracy.

Q11.

In part (a), almost all the students gained the method mark for a relevant graph completely above the x -axis. To earn the accuracy mark, the students needed to indicate 4 on the x-axis, have a cusp there and correct curvature elsewhere. A small number of students did not gain the accuracy mark.

In part (b), the order of operations was crucial here. The great majority of the students recognised that a stretch and a translation were involved and most chose to describe them in that order. However, these students rarely gave the correct vector $\begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$ for the translation and lost one mark as a result. The small proportion of the students who gave the translation first were more likely to gain full marks.

Q12.

(a) Most candidates began by trying to complete the square for both the x terms and the y terms and many were successful. However, several gave the left hand side of the equation for the circle as $(x - 5)^2 + (y - 7)^2$ or $(x + 5)^2 + (y - 7)^2$ and some omitted the plus sign between the squared terms or had a minus sign instead. It was not uncommon to see a negative value on the right hand side of the circle equation and 25 was often seen instead of 49.

(b) (i) This was usually correct, or followed from the candidate's circle equation.

(ii) Provided the candidate's value of k in their circle equation was positive, candidates were given credit for stating $\sqrt{k}$ as the value for the radius.

(c) (i) Candidates usually drew a circle where the centre was approximately in the correct quadrant. Less attention was paid to where the circle cut or touched the axes. It was necessary to show the circle touching the x-axis at (5,0) and cutting the negative y-axis twice. A few candidates neglected to draw any axes at all and scored no marks.

(ii) Those who had found the correct centre and radius usually had the correct coordinates. Some simply wrote pairs of coordinates on their circle but often failed to identify the coordinates of the point that was furthest away from the x-axis.

(d) The majority of candidates knew that the transformation was a "translation" although a few included a further transformation such as a "stretch" and some used incorrect words such as "shift". However, very few stated the correct column vector and $\begin{bmatrix}-6\\7\end{bmatrix}$ and $\begin{bmatrix}-6\\-7\end{bmatrix}$ were very common wrong answers.

Q13.

The sketch of the given exponential curve was usually drawn having the correct intercept at y = 1, but a significant minority of students did not show its correct behaviour for negative x values – the curve should level out close to the x-axis and certainly nearer to the x-axis than the line y = 1. Part (b) was usually answered correctly with almost all showing that they knew how to use logarithms to solve an equation of the form $a^x = b$. The positioning of the minus sign was crucial in part (c) and it was not uncommon to see the incorrect answer f(x) = -9^x.

Q14.

The trapezium rule question in part (a) was generally well answered with the majority of students taking care to include sufficient brackets and rounding their final answers to the required degree of accuracy. In part (b), many students correctly recognised the transformation as a stretch and a majority of these stated its correct direction. However, only the most able students stated the correct scale factor as 2 with most other students incorrectly stating it as either 8 or $\frac{1}{8}$. Although fully correct solutions to part (c) were rare, a significant number of students did gain some credit. The most common approach was to first find an expression for g(x) before substituting the value 4 for x. Other than sign errors, the most common error was to write g(x) = $\sqrt{(x-2)^3+1-0.7}$ or its equivalent. Those who gave the correct expression, $\sqrt{(x-2)^3+1}$ – 0.7, usually went on to score all 3 marks. The other approach, attempting to find and use the original point to transform, was rarely seen.

Q15.

Many students appeared to have covered this part of the specification very superficially. Many students earned no marks for this question because they did not appreciate that the functions were only valid over a restricted domain. Students should be aware that when co-ordinates are requested these should be stated separately as an ordered pair; in this instance credit was given for correct values marked on the axes, but this was a generous concession.

(a) Some students lost the sketch mark because their gradients were wrong or their curves did not reach the x-axis. A small number of students had the y values as $\frac{\pi}{2}$ or 2π. Some students drew the cosine graph or the graph of $\frac{1}{\cos x}$ while a significant proportion did not attempt this part of the question.

(b) All the above issues surfaced again in this part. Some of the students who earned the marks for part (a) carried out wrong transformations.

Q16.

The responses to this question suggested that transformations and curve sketching need more attention.

(a) A common incorrect response was to reflect the curve in the y-axis. Where students correctly reflected in the x-axis, errors occurred in indicating the intersection with the x-axis where answers of 6 and 18 were common. On the y-axis, $-\frac{4}{3}$ and –12 were often seen.

(b) The most common sketch seen was a copy of the original curve with the part of the curve for $x < 0$ reflected back into the first quadrant. Of those who attempted to reflect in the y-axis, some failed to give a recognisable cusp at (0, 4) and many reflected the 'ends' beyond ±6 in the x-axis.

(c) This part was better answered than the two previous parts. The main errors here were giving a stretch factor of –2 or $\frac{1}{2}$, or giving reflection in the x-axis or in the line $y = x$ instead of the y-axis. Students should note that "a horizontal stretch" is not sufficient and the answer should be given as "stretch parallel to the x-axis".

Q17.

In part (a), the factorisation was almost always correct. Perhaps because of a confusion between factors and roots, many solved an equation in this part to give $x = -2$ and $x = 6$. This was considered as further non-contradictory work and was not penalised.

In part (b), the most common score for the graph was 3 marks. A large number of candidates who had the correct x -intercepts drew a distorted parabola, which had a minimum at (0, –12). Some were quite happy to mark the intercepts as (–2, 0) and (6, 0) but then drew a curve with these intercepts equidistant from the origin.

In part (c)(i), many candidates were familiar with the technique of completing the square, although quite a few gave their answer as $(x - 2)^2 - 8$ instead of $(x - 2)^2 - 16$.

In part (c)(ii), most candidates offered the coordinates of the minimum point of a curve, writing (2,–16) as their answer, instead of giving **the minimum value of the expression**. Sadly, many did not return to part (b) to correct their graphs so as to illustrate this correct minimum point.

Part (d) was beyond the understanding of most candidates. A very common incorrect answer was $y = (x + 3)^2 + 2$. Some gained a mark for adding 2 to their quadratic (or for replacing y by $y - 2$) but many also replaced x by $x - 3$ instead of $x + 3$. Quite a number of those who used the correct approach did not actually write down **an equation**, forgetting to include "y = ", and so lost the final mark.

Q18.

In part (a)(i), most candidates recognised that the transformation was a stretch, but the details of the stretch were not as well known. A common error was to state the scale factor as $\frac{1}{6}$ rather than the correct scale factor. Part (a)(ii), finding a suitably simplified expression for g(x), proved to be one of the least well answered parts on the paper. The most common wrong approach was to just subtract 3 from the expression inside the bracket and simplify to get $\left(-2+\frac{x}{3}\right)^6$. The expected error, replacing x by $x + 3$, was also seen quite regularly. More able candidates, who correctly replaced x by $x - 3$, did not always simplify the resulting expression correctly to an acceptable form. In part (b), most candidates successfully applied the binomial theorem as far as finding the binomial coefficients 15 and 20. A significant proportion of these candidates went on to reach the correct answers, though a number of others failed to identify the required coefficients completely or to give $\frac{15}{9}$ in its simplest form. A significant minority could not cope with the additional powers of $\frac{1}{3}$.

Q19.

Very few candidates scored full marks for this question.

Part (a) was well done with only a handful failing to recognise a stretch and most giving the correct direction and scale factor; most who were wrong in one of these were wrong in both. It was essential to use the correct term ‘translate’ for the other, so ‘shift’ did not earn marks and ‘transformation’ was inadequate. A few made an error in the vector but most were correct.

Most candidates were unsuccessful in answering part (b) of the question as their curve was either below the x-axis or continued into the second quadrant. Many placed the curve through e instead of e + 1, and the curvature , for $x < \mathrm{e} + 1$, was often wrong.

Correct answers in part (c)(i) were sometimes spoiled by poor algebra. It was worrying to see the misunderstanding of the log function, with ln(x – e) expanded as ln x – ln e and incorrect subsequent work once a correct answer had been found: $\mathrm{e}^{-1} + \mathrm{e} = 0$ was a common such response.

In part (c)(ii), although the first mark for $x \geq 2\mathrm{e}$ was often earned (though $x \leq 2\mathrm{e}$ was common), the second mark was only gained by a very few candidates, as the part of the inequality concerning e was generally omitted completely.

Q20.

The application of the trapezium rule in part (a), followed by the request for the vector of a translation in part (b), provided a good opportunity for most students to make a good start to this final question. Using log laws in the correct order on $\log(10x^2)$ defeated the average and below average students, even with the answer given. It was quite common to see $\log(10x^2)$ initially written as $2\log(10x)$ and then, with less than convincing logic, the incorrect expression became the right-hand side of the printed result. Identifying and describing the stretch was, as expected, discriminating. Only a small minority identified $\mathrm{f}(ax)$ as $\mathrm{f}(\sqrt{10}\,x)$ here, although many more gained some partial credit. Having this part (c)(ii), following the given result in part (c)(i), was a deliberate connection often ignored by students. Again, in answering part (c)(iii) students could make good use of the printed result, and this was frequently taken up by better students who generally made good progress for the first two or three marks. However, in general, only the most able students could give the correct gradient of OP in the form requested.

Q21.

There were many high marks on this question, although only a minority of students were able to answer part (b) correct.

Students answered part (a) well, with the majority giving a correct modulus graph. The main error was incorrect curvature in the outside branches $x > 3$ and $x < -2$.

In part (b), some students gave an identical response to the modulus graph in part (a), but the most common result was a correct plot for $x \geq 0$ but then the LHS of the curve was reflected in the y-axis to produce a sketch in the first and fourth quadrants.

There were many fully correct solutions to parts (c) and (d). The main errors were the use of the word transformation rather than translation and also the scale factor for the stretch stated as 2 in the x-axis rather than ½ in the y-axis. The common errors in part (d) were (0,5) and (–1,20)

Q22.

The majority of candidates presented a fully correct trapezium rule, working to the required degree of accuracy and so scored full marks. Common errors were normally arithmetical with relatively few candidates failing to score the method mark. In part (b), most candidates indicated that they knew that a multiple of 3 was needed somewhere, although fully correct solutions were

rare. The most common error was to replace x^3 by $\frac{1}{3}x^3$ rather than by $\left(\frac{1}{3}x^3\right)^3$. Another, more serious, error was to multiply the given expression by 3 and write $g(x) = 3\sqrt{27x^3 + 4}$ or some other variation of this error, for example, $g(x) = \sqrt{81x^3 + 12}$.

Q23.

Most candidates replaced $\frac{8}{x^4}$ by $8x^{-4}$ and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a '+c' in their

answer. A very large majority of candidates realised that the value of $\frac{dy}{dx}$ at $x = 1$ was required in
part (b), but a small minority did not appreciated that the value of y was also required. Although many candidates found the equation of the tan gent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for $\frac{dy}{dx}$ to zero and attempted to
solve the resulting equation, although only about half of the candidates obtained the correct coordinates of M. Those who had wrong powers of x in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving
$\frac{d^2y}{dx^2} > 0$ or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate $\frac{8}{x^4}$ correctly, but errors in
the integration of $(x + 3)$ or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the 'Hence', showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the x-axis being a tan gent), 12 (the y value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

Q24.

This question was generally answered very well and full marks were often seen.

In part (a) many correct answers with correct notation were seen but there were many cases of $3 \le f(x) \le -3$ also seen. Where candidates lost a mark it was usually for poor notation.

In part (b)(i) most candidates earned the method mark for swapping x and y but the $\frac{1}{2}$ confused
many when trying to obtain $\cos^{-1}\frac{x}{3}$, with $\cos^{-1}\frac{2x}{3}$ and $\cos^{-1}\frac{x}{6}$ being common errors.

In (b)(ii) having the correct inverse function generally led to the correct answer here; unfortunately repeating incorrect algebra for part (b)(i) sometimes gave the 'right' answer, but obviously without reward.

Part (c)(i) was very well answered.

In part (c)(ii) most candidates achieved the method mark by drawing at least two continuous parts. Unfortunately many lost the final mark, as drawing multiple curves was a common error, as was labelling the x-axis incorrectly.

Part (d) was well done; a few candidates had $\frac{1}{3}$ or $\frac{1}{2}$ as the scale factor and a handful introduced another wrong transformation.

Q25.

In part (a) the coefficient of x, this time being an odd number, caused many problems in the completion of the square; most candidates could not square 2.5 without a calculator and

those who used fractions could not always simplify $7 - \frac{25}{4}$.

In (b)(i) most candidates realised that the vertex was the minimum point but failed to see the link with part (a) and hence were seldom able to write down the correct coordinates. A few were successful using differentiation, but even when the x-coordinate was correct, they often made arithmetic slips in finding the y-coordinate of the vertex.

In part (b)(ii) the equation of the line of symmetry was not usually correct with many writing down

the equation of a curve rather than a straight line; the incorrect wrong equation $y = -\frac{5}{2}$ was seen very often.

In (b)(iii) those with the correct minimum point were usually able to produce a correct sketch, although the value of the y-intercept was sometimes missing from good sketches. Some credit was given to candidates with an incorrect minimum point provided their graph was consistent with this minimum point.

In part (c) only the very able candidates who had answered part (a) correctly earned full marks here. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch (even though this transformation is not within the MPC1 specification). A very common (but incorrect) vector stated by candidates

was $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$, as once again many failed to realise the significance of part (a).

Q26.

Graphs were drawn correctly by the majority of candidates and the correct vector given for the translation. There are still too many candidates, though, who use alternative words for ‘translation’ such as “move” or “shift” which are not accepted.

In part (c)(i) the justification of the transformation of the index equation was more of a challenge even to the high grade candidates. A substantial number of candidates simply replaced 4^x with $(2^x)^2$ which gained no credit as it was basically what they had to show. More were successful in showing that $2^{x+2} = 4Y$ by inserting the intermediate stage $= 2^x \times 2^2$. The final part was only done well by better candidates with many misusing the laws of log s at the outset as illustrated by
‘log 4^x – log $2^{x=2}$ = log 5’. Of those who did solve the printed quadratic equation in Y, equated the values to 2^x and solved for x, where possible, only a minority gave a

conclusion for the "show that" that there was "only one real solution".

Q27.

Part (a) was answered well with most candidates using the trapezium rule with the correct number of strips and values substituted correctly. The most common errors were using degrees instead of radians and incorrect use of brackets.

Most candidates recognised that the required transformation for part (b) was a stretch, but were less sure of the direction and/or scale factor.

In part (c) the most common approach was to attempt to write the given equation in terms of tan x with varying degrees of success. Although some very good solutions were seen, a common wrong method is illustrated by '2tan x = 1, 2x = 0.785, 3.93,….. x = 0.393, 1.97, …..'

Those who correctly reached either $\sin^2 x = \frac{1}{5}$ or $\cos^2 x = \frac{4}{5}$ rarely scored more than one of the four marks because they forgot to consider both the positive and negative roots of these equations.

Q28.

In part (a)(i), although the majority of candidates scored both marks here some rounded to the nearest degree, some truncated, some rounded 14.477 to 14.8, some only considered +4, and some erroneously added –14.5 or 14.5 to 90. A handful took cosec to be $\frac{1}{\tan}$ or $\frac{1}{\cos}$.

In (a)(ii) there were some excellent full solutions here but also some disturbing misunderstanding of notation. Many candidates found it disconcerting to deal with the functions of (2x + 30), for example 2cosec²x – 1 + (2x + 30) = 2 – 7cosec(2x + 30) was seen. They would have been well advised to use a substitution such as Y for (2x + 30). A few missed the brackets around (1 – cosec²Y) thus getting a wrong equation and no further marks; some had this correct but failed to deal with the constants correctly and again could not progress. Sadly the incorrect substitution of cot$Y$ + 1 for cosec Y was also seen on several occasions. Once the correct factors were obtained a few stopped there (particularly those who unfortunately substituted x for 2x + 30). Quite a few handled the –4 solution well but not the $\frac{1}{2}$ as they went on to assume sin (2x + 30) was $\frac{1}{2}$ and not 2.

In part (b) most candidates recognised that the required transformations were a stretch and a translation. The majority started with the stretch and then, wrongly took the translation to be [–30,0] instead of [–15,0]. A few missed the correct term "stretch" and many failed to use the right terminology of "scale factor" $\frac{1}{2}$. A few had the direction of the stretch in the y-axis or the SF as 2 and a few had their translation in the y-direction.

Q29.

In part (a), the slightly unusual form of the initial quadratic caused problems to those who forgot to add 2 after multiplying out the brackets. The first term in the completion of the square was done successfully by most candidates, although the answer $(x - 4)^2 - 3$ was seen almost as often as the correct form. It was unfortunate that an error at this stage was seldom picked up by candidates when they went on to sketch the curve.

In part (b)(i), the sketch was usually of the correct shape but many drew a parabola with the vertex in the wrong quadrant, or the y-intercept was incorrect. The question did ask for the coordinates of the minimum point and this was not always stated by candidates.

In part (b)(ii), a few candidates immediately wrote down the correct equation for the tangent, whereas many felt the need to differentiate in order to find the gradient of the curve at the vertex. Many candidates seemed unaware that the vertex was actually the minimum point and that the tangent at the vertex would be parallel to the line $y = 0$.

In part (c), the more able candidates earned full marks. The term **translation** was required and, although there seemed to be an improvement in candidates' using the correct word, it was still common to see words such as "shift" or "move" being used instead. Others thought that simply writing down a vector was enough. A very commonly stated, but incorrect, vector was $\begin{bmatrix} -4 \\ 1 \end{bmatrix}$.

Q30.

Candidates' sketches generally displayed a thorough understanding of the graph of $y = 2^x$. In part (b)(i) the trapezium rule was frequently applied correctly and a smaller proportion of candidates than normal failed to give their final answer to the stated degree of accuracy. Part (b)(ii) was almost always answered correctly. In part (c), the question asked for a description of a geometrical transformation. Although answers which gave two separate translations were condoned, the penalty for descriptions which involved two different transformations, for example stretch and translation, was severe. However many correct descriptions were presented, which indicates the continued improvement in candidates' performance on this topic. Part (d), as expected, proved to be demanding. The common wrong approach is illustrated by 'log y = log 2^{x+k} + log 3'. Better candidates applied the correct substitution and manipulation to reach $2^k = 5$, but a significant minority of these candidates, after taking logs and using the log law, left their answer in the form $k = \frac{\log 5}{\log 2}$.

Q31.

Parts (a) and (b) gave the opportunity for all to pick up marks, with the majority giving the two decimal place answer required in part (b). Many candidates scored just one of the two marks in part (c). The translation approach was the more popular, but the fact that the vector needed to be $\begin{bmatrix} \frac{3}{4} \\ 0 \end{bmatrix}$ was too subtle for most, as was the scale factor of $\frac{1}{8}$ for those using a stretch approach. Similarly in part (d), needing $4(x - 1)$ rather than $4x - 1$ defeated the majority. However, this caused no undermining of confidence as many answered part e(i) well, with sufficient justification to support the given answer.

It should be noted, however, that answers which simply started with $k = \frac{8 \times 5}{4}$ with no evidence of using correct log laws could not earn credit. In part (e)(ii) most candidates knew they had to solve '$\frac{5}{4} = 2^{4x-3}$' and started by taking logs, but then resorted to calculators rather than establish an exact result (with numbers hinted at in the previous part). The best solution, given by some, was to multiply both sides by '2^3' and then take logs, to the base 10, of '$10 = 2^{4x}$', which led directly to the result.

Q32.

(a) This was well answered by the majority of candidates with many fully correct responses. The main error with the stretch was using the ^-direction with a scale factor of 2 and with the translation the vector often contained +1.

(b) This part was well answered by many candidates.

(c) (i) This was not very well answered by weaker candidates with many just rearranging the expression and then writing down the answer given without any convincing step being shown.

(ii) This was answered better than part c(i) with many correct factorisations seen. The main error was with the rejection of the solution $e^{2x} = -1$.

(d) Some candidates obtained the correct answer and earned full marks. Most candidates made a reasonable attempt at the integrals with reasonable success and most showed enough to earn the B mark for subtraction of their areas. Incorrect limits of 6 and 0 often spoiled what would have been very good attempts.

Q33.

In part (a)(i) the completion of the square was done successfully by most candidates, although occasionally the value 6 was seen instead of 4 for q.

In part (a)(ii) it was necessary to comment on both parts of the expression; $(x + 1)^2 \geq 0$ and hence adding 4 implies that $(x + 1)^2 + 4 > 0$ for all values of x. Because of the word "hence", an argument based on algebra rather than the features of a curve was required; for instance, an answer explaining that $(x + 1)^2 + 4$ has a minimum value of 4 was acceptable, but a statement about the curve having a minimum point at (–1, 4) was not.

In part (b)(i) most candidates were able to write down the correct minimum point. A few chose to use differentiation but sometimes made arithmetic slips in finding the coordinates of the stationary point.

In part (b)(ii) those with the correct minimum point were usually able to produce a correct sketch, although the value of the y-intercept was sometimes missing. Some credit was given to candidates with an incorrect minimum point, usually (1, 4), provided their graph was consistent with this minimum point.

The more able candidates earned full marks in part (c). The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch. A very common (but incorrect) vector stated was $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Q34.

This long calculus question with a final part covering transformations was a relatively high source of marks for many candidates. Most candidates were able to correctly differentiate $2x^{\frac{3}{2}}$, but not all evaluated it correctly at $x = 4$. The majority of candidates showed a good understanding of the method involved in finding the equation of the normal in part (a)(ii), although some others did not use the correct gradient. Although the vast majority of candidates knew the rules for integrating $8x^{\frac{1}{2}}$, there was a significant number of such candidates who did not simplify the coefficient correctly.

It was not uncommon to see either '$\frac{8}{\frac{3}{2}}x^{\frac{3}{2}}$' not simplified or simplified incorrectly to '$12x^{\frac{3}{2}}$'.

Part (b)(ii) was generally answered better than expected, with a majority of candidates applying a correct method for finding the area of the shaded region. Although many candidates realised that the transformation was a 'translation', there were frequently sign errors in the vector, or components were interchanged. It was also disappointing to see a significant increase in the number of descriptions that involved more than one transformation which, as usual, was given no marks; the two transformations quoted were translation and a stretch.

Q35.

Candidates were generally successful in part (a), sometimes aided by the allowances made for (–1, 180).

Part (b) was well answered by many candidates. Common errors were the use of 'shift' and 'transformation' to describe the word 'translation', and the stretch was often given a scale factor of ½ in the x-axis.

In part (c), the shape of the graph was usually correct, but many candidates lost this mark by going into other quadrants. The mark for the labelling of the axis was not so readily earned, as these values were often missing or incorrect.

In part (d)(i), many candidates gave the correct response. The major error was to state that $2y = \cos^{-1}(x - 1)$ for the first step resulting in no marks being awarded.

In part (d)(ii), most candidates at least obtained the method mark. Where candidates correctly differentiated, they went on to achieve full marks.